



March 17, 2026

To Whom It May Concern,

I am writing to recommend Layan Alharbi based on her performance in my *Parallel and Distributed Computing* course. She is a capable and motivated student whose academic preparation and professional skills make her well-suited for both graduate study and employment in computer science-related fields. In particular, she stands out for her research curiosity, initiative, and ability to independently explore advanced technical material beyond course requirements.

Throughout the course, Layan demonstrated consistent engagement in both laboratory exercises and the final team project. She self-learned the C programming language and gained hands-on experience with the tools and programming models used to design parallel and distributed systems. In laboratory sessions, she applied theoretical concepts to practical problems, experimenting with parallel constructs, distributed execution, and performance measurement techniques. These exercises strengthened her understanding of how system design choices translate into real performance outcomes. What distinguished Layan was her consistent effort to go beyond the minimum requirements by reviewing recent technical resources and state-of-the-art approaches to strengthen her solutions.

Layan developed a clear understanding of how algorithmic and system-level design decisions influence performance, scalability, and correctness. She demonstrated the ability not only to recognize sources of performance loss but also to address them through informed design choices, balancing efficiency, resource utilization, and implementation complexity. Importantly, she took into account the characteristics of the underlying hardware when analyzing and improving program performance.

Layan worked on a student-proposed team project to analyze a large-scale dataset containing over one million movie records. The team developed data processing workflows to examine factors influencing movie success, including relationships between revenue, popularity, ratings, and production budgets, representing realistic data-intensive workloads encountered in modern analytics systems. The solution was implemented in serial, shared-memory parallel, and distributed forms, enabling a



systematic evaluation of correctness, scalability, and performance trade-offs across different execution models.

The project emphasized performance tuning and systematic evaluation. Layan contributed to measuring and analyzing speedup and scalability, and to interpreting results by examining parallel and distributed overheads, communication costs, and hardware-related effects. Her contributions demonstrated a strong ability to connect experimental results to core concepts in parallel and distributed computing. She also exceeded project expectations by independently exploring relevant literature, demonstrating strong initiative and academic maturity.

In summary, Layan is a disciplined, adaptable, highly self-directed learner and a strong team player. She communicates technical ideas with clarity, collaborates effectively, and approaches complex, performance-oriented challenges with persistence and professionalism.

I believe Layan is well prepared for graduate study or professional roles in computer science-related fields, and I recommend her for your consideration.

Please do not hesitate to ask me for additional information.

Sincerely,

Dr. Arwa Alsubhi

Assistant Professor
School of Computing
Computer Science and AI Department
Umm AL-Qura University
amsubhi@uqu.edu.sa